Underwater Ultrasound Imaging via Metamaterial lenses Composed of Phononic Crystals

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Abstract: This study introduces the achievement of underwater ultrasound imaging by using metamaterial lenses based on the Luneburg lens, a representative gradient index lens composed of phononic crystals. The metamaterial lenses made of stainless steel were demonstrated numerically and experimentally in an underwater environment.

Luneburg lens[1] is a gradient index (GRIN) lens that focuses waves to the opposite surface of the lens, and the focus does not suffer from aberrations at all incident angles. Transformation optics (TO)[2] have made the lens structures more versatile and have improved focusing capabilities. 2D and 3D metamaterial lenses with GRIN phononic crystals (PCs) have been developed for air and underwater environments[3,4]. Ultrasonic transducers are challenging to detect objects smaller than the wavelength of ultrasonic waves, limiting the resolution. Therefore, to address these challenges, metamaterial lenses capable of achieving subwavelength imaging over a wide field of view (FOV) must be demonstrated[5].

This study introduces GRIN metamaterial lenses based on an aberration-free Luneburg lens using PCs to achieve underwater ultrasound imaging, as shown in **Figure 1**. The spherical and ellipsoidal metamaterial lenses were demonstrated numerically and experimentally; the ellipsoidal lens based on TO reduces the volume. The meta-atoms composing these metamaterial lenses were characterized by dispersion curves, effective material properties, and equifrequency contours (EFCs) using the band structure calculations and the S-parameter retrieval method. **Figure 1(a)** and **(b)** show the experimental setup and schematic diagram to achieve underwater ultrasound imaging, respectively. **Figure 3** shows the numerical and experimental results for imaging in the *yz*- and *xy*-planes of the spherical metamaterial lens. The spherical metamaterial lens has been shown to outperform the ellipsoidal metamaterial lens in beam steering performance, as measured by focal length and full width at half maximum (FWHM). The ellipsoidal metamaterial lens (frequency range of about 60–160 kHz) operate over a higher and wider frequency bandwidth than the spherical metamaterial lens (frequency range of about 30–100 kHz).

In this presentation, we discuss not only the content presented in the paper[6], but also the experimental observation of underwater ultrasound imaging at MHz frequencies with an ellipsoidal metamaterial lens. Furthermore, we will also introduce the quasi-3D nature[7]. That is, we will introduce the contents of underwater ultrasound imaging using metamaterial lenses that our research team has studied so far, and share insights into the field through discussions with researchers participating in this presentation about contents other than those introduced in the paper.

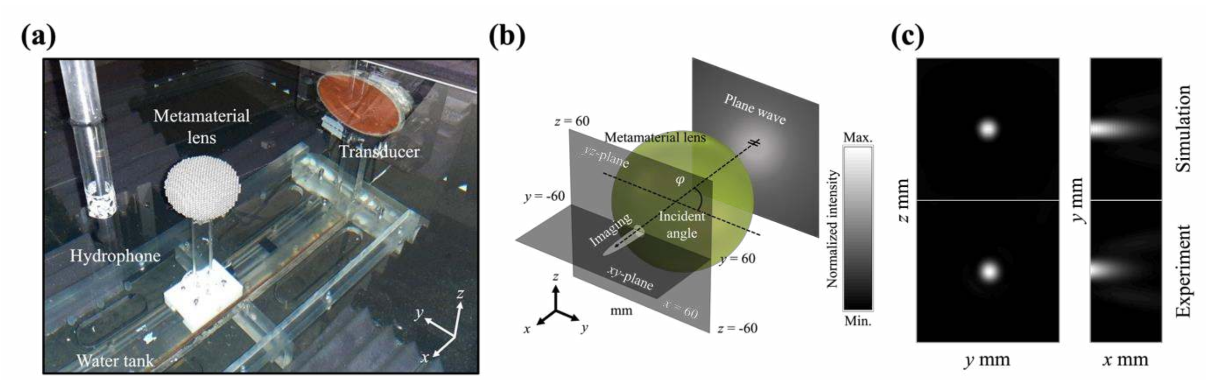

Figure 1 **(a)** Experimental setup and **(b)** schematic diagram for the experimental validation and numerical simulations. **(c)** Underwater ultrasound imaging.

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